

HUJI Applicative Research Report — Agriculture & Food — Week 37, 2026

1 paper(s) · Agriculture & Food · Weekly · Week 37, 2026 · Generated September 07, 2026

■ ■ There is no high-potential applicable research this period — only lower-scoring papers were found. Presenting the two best below.

The latest research from Hebrew University delves into foundational plant physiology, specifically exploring the complex interplay between stomatal opening, vapor pressure deficit (VPD), and transpiration. This work aims to resolve critical questions about how plants regulate water loss, which has profound implications for agricultural efficiency. The findings are poised to enable significant advancements in precision agriculture, offering new pathways for optimizing water use and crop health. We believe this insight will fuel the development of next-generation sustainable farming technologies.

RESEARCHERS & RESEARCH SUBJECTS — BEST AVAILABLE

• Yotam Zait — AgriTech, Software/AI

AGRICULTURE & FOOD — RESEARCH HIGHLIGHTS

1. Stomatal aperture, VPD and transpiration - The chicken or the egg?

10/50

"Unlock significant water savings and optimize crop yields with smarter, AI-driven irrigation systems."

Main researcher: Yotam Zait · yotam.zait@mail.huji.ac.il.

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ABOUT THIS RESEARCH

The research investigates the causal relationship between stomatal opening, vapor pressure deficit (VPD), and plant transpiration rates. It seeks to determine whether stomatal movement drives transpiration or if environmental humidity triggers the opening of these pores.

COMMERCIAL ANGLE

Developing precision sensor technologies or autonomous irrigation systems that can predict and respond to real-time plant gas exchange dynamics to optimize water use efficiency.

COMMERCIALISATION METRICS

Novelty

3/10

The title suggests a mechanistic study of established physiological feedback loops (VPD-stomatal regulation) rather than the introduction of a novel biological pathway or a disruptive technological platform. It appears to be a fundamental refinement of existing plant physiology knowledge rather than groundbreaking scientific innovation.

Commercial Potential

2/10

The title suggests a fundamental mechanistic study in plant physiology rather than a technology-driven application. Without evidence of a platform or tool, it appears to be basic academic research with limited direct commercial translation.

Market Size

2/10

The paper appears to be fundamental plant physiology research focused on basic biological mechanisms rather than a product or technology with a defined commercial application. There is no evidence of a scalable industrial or clinical market being addressed.

Tech Readiness

2/10

The title indicates a fundamental physiological study investigating causal relationships in plant biology, which represents basic scientific research rather than applied technology development. This work is at the earliest stage of knowledge generation (TRL 2) with no immediate path to a commercial product or industrial tool described.

IP Strength

1/10

The title suggests a basic physiological study of plant biology and environmental interactions, which describes natural phenomena that are inherently non-patentable. There is no indication of a novel tool, synthetic construct, or proprietary method that would provide industrial defensibility.

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